Group 9's Final Proposal: Pac-Men

1. Team Introduction

Name	Project Responsibility	Programming Experience
Nguyen Dam Xuan Nguyen	Project Leader, supporting game programmer & UI designer	Extensive understanding and experience in Python, as well as some basic knowledge in Java & C#. Some experience in developing games, websites, …
Le Tri Duc	Main game programmer	Extensive understanding and experience in Python, as well as some basic knowledge in Java
Tran Minh Quang	Idea contributor, supporting game programmer	Advanced understanding and experience in Python, as well as some basic knowledge in C++ and Java.
Nguyen The Nguyen	UI designer	Moderate understanding and experience in Python, basic experience in Java. Some experience in developing websites, etc.
Nguyen Tien Khoi Nguyen	Supporting game programmer	Average understanding and experience in Python, and basic knowledge in Java.

2. Product Description

Title: Pac-Men
Idea: Pac-Man but open for Multi-player
Purpose:

+ Provide an interactive and engaging multiplayer gaming experience.
+ Demonstrate the application of Object-Oriented Programming concepts in Java, especially on Encapsulation and Polymorphism (details on Section 4), as well as data structure concepts like 2D Array grids or Hashmaps.

Objectives:

+ Single-player game (Standard pac-man): 1 player, 3 ghosts. Collect all pellets and special items to score points while dodging ghosts that will follow a certain movement algorithm and kill you on contact.
    - Small pellets: 10 points
    - Power pellets: 50 points; ability to kill ghosts on contact for 10s, providing (200 ^ kill combo) points per ghost killed.
+ Multiplayer **Versus** game (2 players): **Both players play as pac-man**. The first one to be slain will be the loser; If no one dies and both players clear the map, one with a **higher score** wins.

Product type: Downloaded as a Compressed (.zip) file, opened as Application (.exe) file;
GUI and Framework: JavaFX *(Proposed)*

The problem it aims to solve or the need it fulfills:

+ Stress-relievers for amateur players
+ Fulfill human entertainment needs
+ Offers a fresh, exciting multiplayer experience while letting gamers relive nostalgic arcade memories.

3. Target Users

The primary users are casual gamers and retro gaming enthusiasts. Additionally, the game appeals to a broad audience, including children, teenagers, and adults.

Justification:

Its straightforward gameplay requires no complex instructions, making it accessible to all age groups. Retro enthusiasts seek to relive the classic experience, while new players can enjoy its thrilling challenge alone, or with friends.

Benefits:

- Entertainment: Provides engaging and fun gameplay.
- Cognitive Skills: Enhances hand-eye coordination, quick decision-making, and strategic thinking.
- Nostalgia: Recreates a beloved classic for those who grew up with it.
- Accessibility: Easy to learn, free to win.

4. Functional Specifications

4.1 General

a. The system shall support 2-player gameplay on a single device
b. The system shall include two modes: Singleplayer and Multiplayer Versus
c. The system shall define clear win and lose conditions on both modes
d. The system shall handle collision detection between all game objects
e. Player positions shall update in real time

4.2 OOP

a. The system shall implement inheritance by defining a base class for common game objects and extending it for specialized entities (e.g Power Pellets)
b. The system shall implement encapsulation by defining private attributes (score, position, …) within classes and providing controlled access through public methods
c. The system shall implement polymorphism by allowing different objects to respond differently to the same method calls

4.3 Data

a. **Data Structures:** The game shall use 2D Arrays to store the game map, and utilize Queue to store incoming player commands. Additionally, the game shall use Hashmaps to lookup and store game object states
b. **Algorithms:** The game should use efficient pathfinding algorithms, like the Breadth-First Search
c. **Data Storage:** The game shall store player data (e.g. Score) through writing and saving local .txt files

4.4 GUI

a. The system shall provide a graphical interface using JavaFX
b. The system shall display game elements in real time
c. The system shall display player scores and game status
d. The interface shall be intuitive and easy to understand

4.5 Input handling

a. The system shall support simultaneous keyboard inputs
b. The system shall assign different keys for each player

4.6 Constraints

a. The system shall run on a single device
b. The system shall not require internet connectivity

5. Project Timeline

This timeline is devised according to the provided suggestions. However, our team extended the duration for phase 4 due to ongoing Midterm exams (14th to 20th April).

Phase	Name	Description	Start Date	End Date
1	Planning & Analysis	*Proposal Preparation*	March 12th	March 26th
2	Requirements Definition	*Identify game Features*	March 19th	March 26th
3	Design & Prototyping	*Game architecture through diagrams and data structures*	March 26th	April 5th
4	Implementation	*Programming based on design and prototypes*	April 6th	May 9th
5	Refactoring & Feature Development	*Further game development and editing*	April 13th	May 20th
6	Testing & Deployment	*Game testing, debugging, and balancing; Exporting to an publicly accessible application*	May 10th	May 20th
7	Finalization & Documentation	*Optimize, refine, receive feedback*	May 10th	May 30th

6. Appendix

Term Project Team Agreement Form

Course: COMP1020
Group Number: 9
Project Title: Pac-Men
Team Leader: Nguyen Dam Xuan Nguyen
Date: 19/03/2026

Team Member Information

Name	Student ID	Signature / Acknowledgment
Nguyen Dam Xuan Nguyen	V202502804	
Nguyen The Nguyen	V202502897	
Tran Minh Quang	V202502133	
Nguyen Tien Khoi Nguyen	V202502936	
Le Tri Duc	V202502315	

Agreement Terms

By signing this agreement, we, the members of the above project team, acknowledge and agree to the following:

1. **Equal Commitment**: We are each responsible for contributing actively and fairly to the project's success.
2. **Role Assignment**: We will collaboratively determine task assignments based on individual strengths and project requirements.

3. **Communication**: We agree to communicate regularly, respond promptly to messages, and attend scheduled team meetings.
4. **Accountability**: We understand that individual contributions will be evaluated, and non-participation may negatively affect final grades.
5. **Conflict Resolution**: We will attempt to resolve conflicts internally; if unresolved, the issue will be reported to the instructor.
6. **Reporting Non-Participation**: The team leader will report any non-contributing members, and members may also report non-performing leaders.

Acknowledgement: We understand and agree to abide by the terms outlined above. Failure to meet these expectations may result in individual penalties.